

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	AM-PhD
Program Name	PhD in Applied Mathematics
Assessment Year	2024-2025
Program Assessment Coordinator	Michael Pelsmajer
Year of Current Assessment Plan	Year 10 or so. (I inherited this job last year and I don't have all the older documents.)
No Assessment This Year per Plan	☐
LO(s) Assessed	LO1: Students will solve applied problems by applying fundamental concepts of applied real analysis. LO2: Students will solve applied problems by applying fundamental concepts of computational mathematics.
Presented at program authority meeting date: <i>Please attach minutes</i>	Applied Math Graduate Studies Committee 10/8/2025 <i>No minutes provided; most of the meeting has nothing to do with assessments and it's not suitable for public distribution.</i>

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; **{D Does not meet expectations, M Meets expectations, P Proficient}**

LO 1: Problem #2 from the Final Exam in Math 500, our only required applied real analysis course for PhD students, tests three fundamental issues applied real analysis. Details are within the rubric: Rubric-LO1-Fa24-Math500.pdf

LO 2: Problem #2 from the Final Exam in Math 577, our only required applied computational mathematics course for PhD students, tests three fundamental issues applied real analysis. Details are within the rubric: Rubric-LO2-Fa24-Math577.pdf

Results

Artifact samples

Artifact samples are attached.

Attach artifact samples. You can paste them in here if feasible, attach them to the end of this report, or submit them as supplemental files when submitting your reports. For data sets with ≤ 10 students, provide 1 in each category, D,M,P, (unless the category is empty, if so state that). For programs with ≤ 20 , provide 2. For other programs provide 3. Please redact any personally identifiable information, PII (which includes names and A#s). Please convert your documents to PDFs.

Please use the following naming convention for your artifact files: **AY25_CIM-CODE_Name of Artifact.pdf**.

Data table

For each LO and provide a table pasted in from the downloadable [provided template](#), placing each student into the assessment categories (minimally {D, M, P}). You should collect and analyze data linked to student A#s locally, but please redact any personally identifiable information, PII (including names or A#s) in tables presented in this report. Provide a summary graph from the template presenting this distribution. You should also present additional analysis as deemed appropriate by the results seen, and the needs of the program.

Data files attached.

Description of findings

Summarize the discussion that occurred within the program authority when this was presented. Provide a short narrative interpreting the findings. Are there any areas of concern?

Summary: The Graduate Studies Committee of the Applied Mathematics Department reviewed the data and see no issues with MATH 577 and LO1 but expressed concern that the LO2 results confirm that new PhD students are often not prepared academically to take our introductory course, MATH 500.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

We plan to give current and future new graduate students an old undergraduate real analysis (MATH 400) final exam, and if they do not score well, then we will recommend some sort of remediation – to take MATH 400 and delay taking MATH 500, or to do some other kind of review, the structure of which may depend on the number of students needing remediation.

We will also discuss possible ways to help students with LO1, from holding an extra problem session once a week, to entirely reorganizing the analysis requirement for the PhD. However, our options are limited for logistical reasons (few students, tight financials).

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☐ no previous CI plans available

We previously resolved to consider revising the curriculum and emphasis of course topics in MATH 500, with the intention of verifying that the curriculum is still optimal and updating it if needed. No fault was found with the curriculum and no revisions were implemented.

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☐ After consideration and presentation to the program authority, no CI plans are recommended.

See above. Trevor Leslie (the current MATH 500 instructor) will lead all CI efforts for LO1.